

Business Analytics Report

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1. Dataset Overview

Total Rows: 3676

Total Columns: 44

Missing Values: 0

2. Column Information

- customer_id: object
- gender: object
- age: int64
- married: object
- number_of_dependents: int64
- city: object
- zip_code: int64
- latitude: float64
- longitude: float64
- population: int64
- number_of_referrals: int64
- tenure_in_months: int64
- offer: object
- phone_service: object
- avg_monthly_long_distance_charges: float64

3. SQL Analysis Results

Query 1 - Top Categories:

- Rank 1: 9705-IOVQQ
- Rank 2: 9518-XXBXE
- Rank 3: 9402-ORRAH
- Rank 4: 9373-WSLOY
- Rank 5: 9221-OTIVJ

4. Statistical Metrics

- Total Rows: 3676
- Total Columns: 44
- Missing Values: 0

- Duplicate Rows: 0
- age - Total: 188805
- age - Average: 51.36
- age - Median: 52.0
- age - Std Dev: 18.12
- number_of_dependents - Total: 832
- number_of_dependents - Average: 0.23
- number_of_dependents - Median: 0.0
- number_of_dependents - Std Dev: 0.73
- zip_code - Total: 343414758
- zip_code - Average: 93420.77
- zip_code - Median: 93444.5
- zip_code - Std Dev: 1878.28
- latitude - Total: 132748.38
- latitude - Average: 36.11
- latitude - Median: 35.86
- latitude - Std Dev: 2.46

5. AI Analysis & Recommendations

****1. Business Domain:****

The business dataset appears to belong to a telecommunications company that provides internet and phone services to its customers. The dataset likely contains information about customer demographics, service usage, and billing data.

****2. Key Findings:****

1. ****Average Customer Age:**** The average age of customers is 51.36 years, with a median age of 52.0 years. This suggests that the target market for the company is likely middle-aged individuals.
2. ****Dependents:**** The average number of dependents per customer is 0.23, indicating that most customers do not have dependents. This may impact the company's marketing strategies and service offerings.
3. ****Zip Code Clustering:**** The standard deviation of zip codes is 1878.28, indicating a moderate level of spread across different regions. The average zip code is 93420.77, with a median of 93444.5. This suggests that the company has customers spread across various regions, but with a higher concentration in specific areas.

****3. Business Risks:****

1. ****Churn:**** The dataset contains a "churn_label" column, which indicates customers who have churned (i.e., left the service provider). The churn score is 91.0, suggesting that a significant portion of customers are at risk of leaving the company.
2. ****Competition:**** Some customers have cited "Competitor offered more data" as the reason for churning. This highlights the company's need to review its pricing plans and offerings to remain competitive in the market.
3. ****Limited Service Range:**** Some customers have expressed dissatisfaction with the limited range of services offered by the company. This highlights the need for the company to assess its service offerings and consider expanding its portfolio to meet customer needs.

****4. Recommendations:****

1. ****Targeted Marketing:**** Develop targeted marketing campaigns directed at middle-aged individuals, focusing on services that cater to their needs.
2. ****Service Offerings:**** Review and expand the company's service offerings to cater to a broader range of customers,

including those with dependents.

3. **Pricing Strategy:** Re-examine pricing plans to ensure they remain competitive in the market, particularly in areas where customers are citing competitors as the reason for churning.

4. **Customer Feedback Mechanisms:** Establish robust customer feedback mechanisms to gather insights on customer satisfaction and areas for improvement.